# Aashan Javed

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## Research Interests

**Calibrated uncertainty for frozen foundation models under distribution shift**, applied to climate and urban heat. My work asks whether published reliability guarantees survive the shift a deployed model actually meets, and so far the answer has been no.

## Education

**National University of Computer and Emerging Sciences (FAST-NUCES)** 2020 – 2024
*Bachelor of Science in Computer Science* *Islamabad, Pakistan*
- Awarded the Gold Medal and a place on the Rector's List of Honour for the seventh semester. Batch rank 57 of 285, the top 20 percent.

## Research papers

**How Fine Can Your Groups Get? The Breakdown of Plug-in Worst-Group Accuracy**
A. Javed
*Under review, AISTATS 2027.*
Worst-group accuracy is the standard criterion for certifying group robustness, and it is almost always estimated by the plug-in minimum of per-group accuracies. That estimator's error is **model-dependent**: it vanishes when one group is clearly worst and is largest when group accuracies are near-tied. The bias therefore does not cancel when two models are compared, so a ranking can invert on the estimator alone.

**When Does a Frozen Time-Series Foundation Model's Uncertainty Fail, and What Actually Fixes It?**
A. Javed
*Preprint, 2026; manuscript under revision.* code
The natural rolling-origin implementation of this evaluation leaks its own future into calibration and overstates the apparent recovery, so every result is reported lagging feedback by the full horizon.

## Research Experience

**Independent Researcher** 2025 – Present
*Calibrated uncertainty under distribution shift* *Remote*
- Designed and ran the experiments behind my research end to end: problem selection, literature gap verification, data acquisition, implementation, evaluation and writing.
- Built a model-agnostic, unit-tested conformal prediction library (split-conformal, CQR, Mondrian, adaptive conformal inference) reused across every study.
- Ran an adversarial audit of my own results and found that a masking flag I believed was off had never reached the frozen encoder, so two calls on the same input disagreed. **I stopped the affected work rather than publish on it.** Reproducing a frozen encoder's output before trusting it, and reporting a falsified prediction rather than discarding it, are now fixed steps in how I work.

## Professional Experience

**Founder and CEO** Jan 2024 – Present
*Valfirst AI (Valfirst Software (SMC-Private) Ltd), side venture* *Pakistan / Remote*
- Founded and lead **Valfirst AI**, which builds and operates autonomous AI agents that run real business workflows end to end for operations-heavy small and mid-sized companies, not chatbots but digital workers that respond to leads, qualify customers, and handle repetitive processes.
- Built and deployed **11 live, interactive AI agents** across industries, demoable with no login at valfirstai.com; agents respond to and qualify a lead in seconds, integrate into a client's tools, and report via a monthly scorecard.
- In pilot: running the first paid project, an agent system that manages the digital media operations for Proton Studios.

**Machine Learning Engineer (R&D)** Jan 2025 – Present
*FortyGuard, urban heat and climate intelligence* *Abu Dhabi, UAE (Remote)*
- Built the **Large Temperature Model (LTM)**, forecasting street-level (2 m) air temperature at 10 m resolution across the United States.
- Led a collaboration with **NVIDIA** on a custom **CorrDiff diffusion-based downscaling model**, a 255× spatial amplification (16× finer per axis) taking a 25 km grid to roughly 1.5 km, for high-fidelity urban climate modelling (San Jose region).
- Currently building FortyGuard's geospatial agents. Own multi-source geospatial pipelines (**10M+ observations/day**) and model evaluation and drift monitoring in production, where the gap between reported accuracy and deployed reliability became concrete.

**Software Engineering Intern** Jun 2024 – Dec 2024
*FortyGuard* *Abu Dhabi, UAE (Remote)*
- Built dashboards over 100K+ records/day and the data plumbing the temperature modelling later ran on.

**Machine Learning Engineer Intern** Jun 2023 – Sep 2023
*Payactiv* *Islamabad, Pakistan*
- Built gradient-boosted risk scoring over 1M+ transaction records, where calibrated abstention mattered more than raw accuracy.

## Community & Service

**FortyGuard Global AI Hackathon '26** 2026
*Mentor* *Remote*
- Served as a mentor for a global climate-tech hackathon and led the session "Build with Heat Intelligence" on 19 August 2026, on working with temperature and heat data in applied projects.

**Climate Change AI (CCAI)** 2025 – Present
*Community member and volunteer* *Remote*
- Active member of the Climate Change AI community; collaborate on and volunteer at its research events.

**Alkhidmat Foundation Pakistan** 2022
*Digital Volunteer, Flood Relief* *Pakistan (Remote)*
- Supported digital outreach and awareness during the 2022 flood emergency, amplifying relief and donation drives to connect support with affected communities. Recognised with a Certificate of Appreciation.

**FAST Community Service Society** 2021 – 2023
*Head of Information and Communication* *Islamabad, Pakistan*
- Led communications and volunteer coordination for FAST's student-run community service society, running its social media, announcements, and event promotion, and mobilising volunteers across charity drives and student outreach activities.

**Google Developer Groups Islamabad, Youth For Pakistan, peer mentoring** 2021 – Present
*Community volunteer, campus ambassador, peer mentor* *Islamabad, Pakistan*
- Helped plan and run developer meetups and workshops, represented Youth For Pakistan on campus for over two years, and continue to mentor students on university admissions and field selection.

## Technical Skills

**Uncertainty & Evaluation**: Conformal Prediction (split, CQR, Mondrian, adaptive), Calibration Assessment, Coverage & Interval-Width Analysis, Distribution-Shift Evaluation, Leakage-Controlled Experimental Design

**Machine Learning**: Python, PyTorch, scikit-learn, XGBoost, Foundation-Model Representation Probing, Time-Series Forecasting, Spatiotemporal Modelling

**Geospatial**: Sentinel-2, Landsat C2, Microsoft Planetary Computer, rasterio, Cloud-Optimised GeoTIFF, Local Climate Zones

**Research Engineering**: Reproducible Pipelines, Docker, CI/CD, LaTeX, Git

**Languages**: English (**IELTS Academic 7.5**), Urdu (native)